# NDC Booking Lifecycle

Full Reference — SeatMap, FareCheck, Hold, FareConfirmAfterHold, Book, BookAfterHold, RetrieveBooking

Requests, responses, field definitions and worked samples — AirArabia V2

## 1. Overview

This document is the single reference for every V2 endpoint touching seat selection and pricing, end to end: what each request and response actually contains, field by field, plus a worked sample of both for every step. All samples share one continuous scenario so the same PNR, passengers and seats can be traced through the whole lifecycle.

| Field | Type | Description / purpose |
|---|---|---|
| `GET SeatMap` | *SeatMapRequestDto / SeatMapResponseDto* | Seat map for a shopped, not-yet-booked offer. |
| `FareCheck` | *FareCheckRequestDto / FareCheckResponseDto* | Re-prices the shopped offer with ancillary/seat selections applied, before any PNR exists. |
| `Hold` | *HoldRequestDto / HoldResponseDto* | Creates a PNR on hold (no ticketing) with the selected seats/ancillaries. |
| `FareConfirmAfterHold` | *FareConfirmAfterHoldRequestDto / FareConfirmAfterHoldResponseDto* | Re-confirms price on an already-held PNR before final booking. |
| `Book` | *FlightBookRequestDto / FlightBookResponseDto* | Books and tickets directly, without an intervening hold. |
| `BookAfterHold` | *BookAfterHoldRequestDto / BookAfterHoldResponseDto* | Converts an existing held PNR into a ticketed booking. |
| `RetrieveBooking` | *RetrieveBookingRequestDto / RetrieveBookingResponseDto* | Looks up an existing PNR at any later point. |

## 2. GET SeatMap

### 2.1 Request — SeatMapRequestDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier` | *enum* | Which supplier/GDS to query. |
| `credentialsSelector` | *enum* | Which credential set to use for that supplier. |
| `responseId` | *Guid* | Correlates back to the cached pricing/search response this offer came from. |
| `selectedOfferId` | *string* | The offer whose seat map is being requested. |

Requests the seat map for a priced-but-not-yet-booked offer, identified by the same ResponseId/SelectedOfferId pair returned from search/pricing.2.2 Request — PostSeatMapRequestDto (after hold)

Same seat map, but scoped to an existing held PNR instead of a shopped offer. Extends OrderBaseRequestDto (§5.1 shows the full shared shape) with no extra fields — the PNR/bookingToken alone is enough to look the seat map up.

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / credentialsSelector` | *enum* | Same meaning as above. |
| `pnr / gdsPNR / bookingToken` | *string?* | Identifies the held booking whose seat map is being requested — typically just pnr is needed. |
| `surName` | *string?* | Passenger surname, required by some suppliers to authorize a PNR lookup. |

## 2.3 Response — top level

| Field | Type | Description / purpose |
|---|---|---|
| `supplier` | *enum (string)* | Airline/GDS supplier that produced this seat map. |
| `flightSegments` | *Dictionary<segmentId, SegmentType>* | Flight segment detail for every segment referenced by seatMapDetails. |
| `passengersDetails` | *Dictionary<paxId, PassengerDetails>* | Passenger type, name, birth date and travel document for every passenger. |
| `seatMapDetails` | *Dictionary<segmentId, SeatMapDetails>* | The seat map itself — cabin layout and every row/cell — one entry per segment. |
| `passengersSeatMapAvailability` | *Dictionary<paxId, Dictionary<segmentId, List<string>>>* | Seat codes each passenger can actually select, narrowed from the full grid. |

### 2.3.1 SeatMapDetails

| Field | Type | Description / purpose |
|---|---|---|
| `cabinLayoutsByDeck` | *Dictionary<DeckType, CabinLayoutType>* | Column layout for each deck present on the aircraft. Single-deck aircraft have one entry; wide-bodies with an upper deck have one entry per deck. |
| `rows` | *List<RowType>* | Every row of the cabin, flattened across all decks; each row states which deck it's on. |

### 2.3.2 RowType

| Field | Type | Description / purpose |
|---|---|---|
| `rowNumber` | *int* | Physical row number as printed on the aircraft. |
| `deck` | *enum DeckType* | Main / Upper / Lower / Unknown. |
| `cells` | *List<CellType>* | Every position across the row, left to right. |

### 2.3.3 CellType

| Field | Type | Description / purpose |
|---|---|---|
| `cellCategory` | *enum CellCategory* | Seat / Aisle / Lavatory / Other. |

| Field | Type | Description / purpose |
|---|---|---|
| `column` | *string?* | Column letter, e.g. "A"; null for non-seat cells. |
| `cellDetails` | *CellDetails?* | Populated only when cellCategory is Seat. |

### 2.3.4 CellDetails

| Field | Type | Description / purpose |
|---|---|---|
| `cellID` | *string* | Supplier's internal identifier for this seat. |
| `cellCode` | *string* | Human-facing seat code, e.g. "12A". |
| `cellAvailability` | *enum CellAvailability* | Blocked / Occupied / AvailableFree / AvailablePaid / AvailableAferReservation. |
| `name` | *string?* | Supplier's display name for the seat. |
| `description` | *string?* | Free-text description. |
| `seatPriceDetails` | *SeatPriceDetails?* | Single price breakdown (legacy, kept for backward compatibility). |
| `seatPriceDetailsByPassenger` | *Dictionary<paxId, SeatPriceDetails>?* | Price breakdown per passenger for this same seat. |
| `cellCharacteristics` | *List<CellCharacteristicType>* | Window/aisle/exit-row etc., as standardized PADIS codes. |

### 2.3.5 CellCharacteristicType

| Field | Type | Description / purpose |
|---|---|---|
| `padisCode` | *enum PadisCode* | IATA PADIS standard code, e.g. W (window), A (aisle), E (exit row). Serializes as its short code string. |
| `description` | *string* | Human-readable description; a plain settable field, populated from the shared PadisSeatCharacteristicHelper lookup (§9.6). |

### 2.3.6 CabinLayoutType

| Field | Type | Description / purpose |
|---|---|---|
| `columns` | *List<string?>* | Column letters left-to-right for one deck; null marks an aisle gap, e.g. ["A","B","C",null,"D","E","F"]. |

## 2.4 Sample — GET SeatMap request

```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "responseId": "3f9a1b2c-7d4e-4a6f-9c2b-8e1d5f0a6b31",
  "selectedOfferId": "OFR-AA-20260726-0091"
}
```

## 2.6 Sample — GET SeatMap response

Row 12 shown (the exit row, with priced seats); full response including ordinary rows 11 and 13 is in SeatMap-Response-Sample.json alongside this document.

```
{
  "supplier": "AirArabia",
  "flightSegments": { "SEG1": { "origin": "DXB", "destination": "CAI",
      "operatingCarrierCode": "G9", "operatingFlightNumber": "402", "equipment": "32A", ... } },
  "passengersDetails": {
    "PAX1": { "passengerTypeCode": "ADT", "name": { "first": "Ahmed", "last": "Al Sayed" }, ... },
    "PAX2": { "passengerTypeCode": "CHD", "name": { "first": "Layla", "last": "Al Sayed" }, ... }
  },
  "seatMapDetails": { "SEG1": {
    "cabinLayoutsByDeck": { "Main": { "columns": ["A","B","C",null,"D","E","F"] } },
    "rows": [
      { "rowNumber": 12, "deck": "Main", "cells": [
        { "cellCategory": "Seat", "column": "A", "cellDetails": {
            "cellID": "SEG1-12A", "cellCode": "12A", "cellAvailability": "AvailablePaid",
            "description": "Window seat, Exit and emergency exit",
            "seatPriceDetails": { "totalPrice": { "amount": 25.00, "currency": "AED" }, ... },
            "seatPriceDetailsByPassenger": {
              "PAX1": { "totalPrice": { "amount": 25.00, "currency": "AED" }, ... },
              "PAX2": { "totalPrice": { "amount": 25.00, "currency": "AED" }, ... }
            },
            "cellCharacteristics": [
              { "padisCode": "W", "description": "Window seat" },
              { "padisCode": "E", "description": "Exit and emergency exit" }
            ]
        }},
        { "cellCategory": "Seat", "column": "B", "cellDetails": {
            "cellID": "SEG1-12B", "cellCode": "12B", "cellAvailability": "AvailablePaid",
            "description": "Aisle seat, Exit and emergency exit",
            "seatPriceDetails": { "totalPrice": { "amount": 15.00, "currency": "AED" }, ... },
            "cellCharacteristics": [
              { "padisCode": "A", "description": "Aisle seat" },
              { "padisCode": "E", "description": "Exit and emergency exit" }
            ]
        }},
        ... columns C, D, E, F ...
      ]}
    ]
  }},
  "passengersSeatMapAvailability": {
    "PAX1": { "SEG1": ["11A","11B","11D","11E","11F","12A","12B","12C","12D","12E","12F","13A", ...] },
    "PAX2": { "SEG1": ["11A","11B","11D","11E","11F","12A","12B","12C","12D","12E","12F","13A", ...] }
  }
}
```

# 3. FareCheck

Re-prices a shopped offer with the client's ancillary and seat selections applied. Runs before any PNR exists, so seat data comes from the request's own selections rather than a confirmed booking.

## 3.1 Request — FareCheckRequestDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / credentialsSelector` | *enum* | Supplier and credential set to price against. |
| `responseId` | *Guid* | Correlates to the cached search/pricing response. |
| `selectedOfferId` | *string* | The offer being re-priced. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, SetPassengerAncillary>?* | Baggage/meal/seat selections per passenger (§9.4/9.5) — seat selections are tagged additionalServiceCategory = "Seat". |

| Field | Type | Description / purpose |
|---|---|---|
| `passengersSelectedSeats` | *Dictionary<paxId, SetPassengerSeatSelection>?* | Dedicated seat selection per passenger/segment (§9.6/9.7). |

## 3.2 Response — FareCheckResponseDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / journeys / flightSegments / priceClasses / baggageDetails` | — | Shared base fields, same meaning as every other response in this document. |
| `responseId` | *Guid* | Echoes the request's responseId. |
| `passengersDetails` | *Dictionary<paxId, PassengerDetails>* | Passenger type/name/birth date/document. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, PassengerSelectedAncillary>* | Echoes the resolved ancillary selections, with decoded price/description. |
| `selectedOffer` | *FareCheckOrderType* | The re-priced offer — see §3.2.1 for the per-passenger breakdown. |

### 3.2.1 selectedOffer.passengerFareBreakdown — seat price fields

| Field | Type | Description / purpose |
|---|---|---|
| `passengerAncillaries / paxTotalAncillaryAmount` | — | Unfiltered, as before — still includes the seat cost, exactly as it always has. |
| `paxTotalSeatAmount` | *Money?* | Sum of just this passenger's Seat-category selections. |
| `passengerSeats` | *ICollection<BookedSeatDetails>?* | One entry per seat-category selection — cellCode/name are the ancillary code, cellCharacteristics is always empty here (no seat-map grid to pull characteristics from at this stage). |

## 3.3 Sample — FareCheck request

```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "responseId": "3f9a1b2c-7d4e-4a6f-9c2b-8e1d5f0a6b31",
  "selectedOfferId": "OFR-AA-20260726-0091",
  "passengersSelectedAncillaries": {
    "PAX1": {
      "setSegmentsAncillaries": {
        "SEG1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      },
      "setJourneysAncillaries": {
        "J1": [
```

```
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      }
    },
    "PAX2": {
      "setSegmentsAncillaries": {},
      "setJourneysAncillaries": {}
    }
  },
  "passengersSelectedSeats": {
    "PAX1": {
      "setSegmentsSeats": { "SEG1": { "cellCode": "12A", "cellId": "SEG1-12A" } }
    },
    "PAX2": {
      "setSegmentsSeats": { "SEG1": { "cellCode": "12B", "cellId": "SEG1-12B" } }
    }
  }
}
```

### 3.4 Sample — FareCheck response

Full response in FareCheck-Response-Sample.json; passengerFareBreakdown excerpt below.

```
"passengerFareBreakdown": [
  {
    "passengerRefId": "PAX1",
    "paxTotalAmount": { "amount": 545.00, "currency": "AED" },
    "passengerAncillaries": [
      { "ancillaryCode": "20KG", "amount": { "amount": 45.00, "currency": "AED" } },
      { "ancillaryCode": "12A", "amount": { "amount": 25.00, "currency": "AED" } }
    ],
    "paxTotalAncillaryAmount": { "amount": 70.00, "currency": "AED" },
    "paxTotalSeatAmount": { "amount": 25.00, "currency": "AED" },
    "passengerSeats": [
      { "cellCode": "12A", "name": "12A", "description": "Window seat, Exit and emergency exit",
        "price": { "amount": 25.00, "currency": "AED" }, "cellCharacteristics": [] }
    ],
    ...
  },
  { "passengerRefId": "PAX2", "paxTotalAmount": { "amount": 385.00, "currency": "AED" },
    "paxTotalSeatAmount": { "amount": 15.00, "currency": "AED" }, ... }
]
```

## 4. Hold

Creates a PNR on hold, with no ticketing, from a re-priced (or directly shopped) offer.

### 4.1 Request — HoldRequestDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / credentialsSelector` | *enum* | Same meaning as FareCheck. |
| `responseId` | *Guid* | Correlates to the cached pricing response. |
| `selectedOfferId` | *string* | The offer being held. |
| `selectedBundles` | *ICollection<SelectedBundleType>?* | Fare bundle selections, if any. |
| `bspCommissionType` | *BspCommission?* | BSP commission instructions, when applicable. |

| Field | Type | Description / purpose |
|---|---|---|
| `passengersSelectedAncillaries` | *Dictionary<paxId, SetPassengerAncillary>?* | Baggage/meal/seat selections — same shape as FareCheck. |
| `passengersSelectedSeats` | *Dictionary<paxId, SetPassengerSeatSelection>?* | Dedicated seat selection, keyed paxId → segmentId → { cellCode, cellId } (§9.6/9.7). |
| `sendEmailNotification` | *bool?* | Whether the supplier should send its own confirmation email. |

## 4.2 Response — HoldResponseDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / pnr / gdsPNR / bookingToken` | — | PNR is now populated — this is where a PNR first exists in the lifecycle. |
| `voidCapability` | *enum* | Whether/how this hold can be voided. |
| `holdExpirationDate` | *DateTimeOffset?* | When the hold expires if not converted. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, PassengerSelectedAncillary>* | Resolved ancillary selections. |
| `passengersBookedSeats` | *Dictionary<paxId, Dictionary<segmentId, BookedSeatDetails>>* | Seat(s) actually assigned on the new hold — now sourced from the confirmed PNR, not the request. |
| `selectedOffer` | *HoldOrderType* | Same passengerFareBreakdown shape as FareCheck, but paxTotalSeatAmount/passengerSeats are now sourced from the confirmed hold. |
| `journeys / flightSegments / priceClasses / baggageDetails / passengersList / contacts / isBspCommissionApplied` | — | Shared booking-response base fields. |

## 4.3 Sample — Hold request

```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "responseId": "9c4e2a71-1b3d-4f8a-8e6c-2d7f0a5b9c14",
  "selectedOfferId": "OFR-AA-20260726-0091",
  "selectedBundles": [],
  "bspCommissionType": null,
  "passengersSelectedAncillaries": {
    "PAX1": {
      "setSegmentsAncillaries": {
        "SEG1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      },
      "setJourneysAncillaries": {
        "J1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      }
```

```
    },
    "PAX2": {
      "setSegmentsAncillaries": {},
      "setJourneysAncillaries": {}
    }
  },
  "passengersSelectedSeats": {
    "PAX1": {
      "setSegmentsSeats": {
        "SEG1": { "cellCode": "12A", "cellId": "SEG1-12A" }
      }
    },
    "PAX2": {
      "setSegmentsSeats": {
        "SEG1": { "cellCode": "12B", "cellId": "SEG1-12B" }
      }
    }
  },
  "sendEmailNotification": true
}
```

### 4.4 Sample — Hold response

Full response in Hold-Response-Sample.json; key seat fields below.

```
"pnr": "6XJ4KP",
"holdExpirationDate": "2026-07-27T09:15:00Z",
"passengersBookedSeats": {
  "PAX1": { "SEG1": { "cellCode": "12A", "description": "Window seat, Exit and emergency exit",
                      "price": { "amount": 25.00, "currency": "AED" },
                      "cellCharacteristics": [ { "padisCode": "W", ... }, { "padisCode": "E", ... } ] }
},
  "PAX2": { "SEG1": { "cellCode": "12B", "description": "Aisle seat, Exit and emergency exit",
                      "price": { "amount": 15.00, "currency": "AED" }, ... } }
},
"selectedOffer": {
  "passengerFareBreakdown": [
    { "passengerRefId": "PAX1", "paxTotalSeatAmount": { "amount": 25.00, "currency": "AED" },
      "passengerSeats": [ { "cellCode": "12A", ... } ], ... },
    { "passengerRefId": "PAX2", "paxTotalSeatAmount": { "amount": 15.00, "currency": "AED" }, ... }
  ]
}
```

## 5. FareConfirmAfterHold

Re-confirms price on an already-held PNR before converting it to a ticketed booking. AirArabia implements this by calling the RetrieveBooking mapper for the held PNR and copying its result across, so its seat data is always identical to what RetrieveBooking would show for the same PNR at the same moment.

### 5.1 Request — FareConfirmAfterHoldRequestDto (OrderBaseRequestDto)

This shared base is also used, unmodified, by PostSeatMap, BookAfterHold and RetrieveBooking requests — referenced from here as §5.1 throughout this document.

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / credentialsSelector` | *enum* | Supplier and credential set. |
| `pnr` | *string?* | The held PNR to re-price. Max length 9. |
| `gdsPNR` | *string?* | GDS-level record locator, for GDS-aggregated suppliers. |

| Field | Type | Description / purpose |
|---|---|---|
| `bookingToken` | *string?* | Alternative to PNR for identifying the held offer. |
| `surName` | *string?* | Passenger surname, required by some suppliers to authorize the lookup. |
| `sendEmailNotification` | *bool?* | Whether the supplier should send its own notification email. |

### 5.2 Response — FareConfirmAfterHoldResponseDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / journeys / flightSegments / priceClasses / baggageDetails` | — | Shared base fields. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, PassengerSelectedAncillary>* | Carried over from the RetrieveBooking call for this PNR. |
| `passengersBookedSeats` | *Dictionary<paxId, Dictionary<segmentId, BookedSeatDetails>>* | Also carried over from RetrieveBooking — same shape/meaning as Hold/Book/BookAfterHold/RetrieveBooking. |
| `selectedOffer` | *FareConfirmAfterHoldOrderType* | Same passengerFareBreakdown shape as every other OrderType — paxTotalSeatAmount/passengerSeats inherited automatically since the whole breakdown is copied from RetrieveBooking. |

### 5.3 Sample — FareConfirmAfterHold request


```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "pnr": "6XJ4KP",
  "gdsPNR": null,
  "bookingToken": null,
  "surName": "Al Sayed",
  "sendEmailNotification": null
}
```


### 5.4 Sample — FareConfirmAfterHold response

Full response in FareConfirmAfterHold-Response-Sample.json — identical seat data to Hold/RetrieveBooking for PNR 6XJ4KP.


```
"passengersBookedSeats": {
  "PAX1": { "SEG1": { "cellCode": "12A", ... "price": { "amount": 25.00, "currency": "AED" } } },
  "PAX2": { "SEG1": { "cellCode": "12B", ... "price": { "amount": 15.00, "currency": "AED" } } }
},
"selectedOffer": { "passengerFareBreakdown": [
  { "passengerRefId": "PAX1", "paxTotalSeatAmount": { "amount": 25.00, "currency": "AED" }, ... },
  { "passengerRefId": "PAX2", "paxTotalSeatAmount": { "amount": 15.00, "currency": "AED" }, ... }
]}
```


## 6. Book

Books and tickets directly from a priced offer, without an intervening hold. Uses the same FlightBookBaseRequestDto as Hold (§4.1), so the request shape is identical.

## 6.1 Request — FlightBookRequestDto

Same fields as HoldRequestDto (§4.1): supplier, credentialsSelector, responseId, selectedOfferId, selectedBundles, bspCommissionType, passengersSelectedAncillaries, passengersSelectedSeats, sendEmailNotification.

## 6.2 Response — FlightBookResponseDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / pnr / gdsPNR / bookingToken` | — | PNR is now ticketed. |
| `voidCapability` | *enum* | Whether/how the ticket(s) can be voided. |
| `creationDate / ticketingDate` | *DateTime?* | When the PNR was created and when tickets were issued. |
| `tickets` | *Dictionary<paxId, TicketType>* | Ticket number and status per passenger. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, PassengerSelectedAncillary>* | Confirmed ancillary selections — read from the confirmed PNR, not echoed from the request. |
| `passengersBookedSeats` | *Dictionary<paxId, Dictionary<segmentId, BookedSeatDetails>>* | Confirmed seat assignment, same shape throughout the lifecycle. |
| `selectedOffer` | *BookOrderType* | Same passengerFareBreakdown shape as Hold, sourced from the confirmed PNR. |

> *Historical note: AirArabia's plain Book response previously echoed the requested price instead of the confirmed PNR price (a bug, since fixed) — it now reads confirmed data exactly like Hold/BookAfterHold/RetrieveBooking always did.*

## 6.3 Sample — Book request

```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "responseId": "3f9a1b2c-7d4e-4a6f-9c2b-8e1d5f0a6b31",
  "selectedOfferId": "OFR-AA-20260726-0091",
  "selectedBundles": [],
  "bspCommissionType": null,
  "passengersSelectedAncillaries": {
    "PAX1": {
      "setSegmentsAncillaries": {
        "SEG1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      },
      "setJourneysAncillaries": {
        "J1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      }
    },
    "PAX2": {
      "setSegmentsAncillaries": {},
      "setJourneysAncillaries": {}
    }
  },
  "passengersSelectedSeats": {
    "PAX1": {
```

```
      "setSegmentsSeats": {
        "SEG1": { "cellCode": "12A", "cellId": "SEG1-12A" }
      }
    },
    "PAX2": {
      "setSegmentsSeats": {
        "SEG1": { "cellCode": "12B", "cellId": "SEG1-12B" }
      }
    }
  },
  "sendEmailNotification": true
}
```

### 6.4 Sample — Book response

Full response in Book-Response-Sample.json — same seat/price data as Hold, now alongside issued ticket numbers.

```
"tickets": {
  "PAX1": { "ticketStatus": "Issued", "ticketNumber": "1416512345", "carrierCode": "G9" },
  "PAX2": { "ticketStatus": "Issued", "ticketNumber": "1416512346", "carrierCode": "G9" }
},
"passengersBookedSeats": {
  "PAX1": { "SEG1": { "cellCode": "12A", "price": { "amount": 25.00, "currency": "AED" }, ... } },
  "PAX2": { "SEG1": { "cellCode": "12B", "price": { "amount": 15.00, "currency": "AED" }, ... } }
}
```

## 7. BookAfterHold

Converts an existing held PNR into a ticketed booking.

### 7.1 Request — BookAfterHoldRequestDto (OrderBaseRequestDto + …)

| Field | Type | Description / purpose |
|---|---|---|
| `supplier` / `credentialsSelector` / `pnr` / `gdsPNR` / `bookingToken` / `surName` / `sendEmailNotification` | — | Shared OrderBaseRequestDto fields, see §5.1. |
| `bspCommissionType` | *BspCommission?* | BSP commission instructions, when applicable. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, SetPassengerAncillary>?* | Lets ancillaries be added/changed at ticketing time. |

> *BookAfterHoldRequestDto does not currently have a passengersSelectedSeats field the way Book/Hold/FareCheck do — seats are assumed already locked in from the Hold step and aren't re-submitted here. Worth flagging if you want seat changes to be possible at this stage too.*

### 7.2 Response — BookAfterHoldResponseDto

Same field set as Book (§6.2): voidCapability, creationDate/ticketingDate, tickets, passengersSelectedAncillaries, passengersBookedSeats, selectedOffer (BookAfterHoldOrderType).

### 7.3 Sample — BookAfterHold request

```
{
  "supplier": "AirArabia",
```

```
  "credentialsSelector": "Default",
  "pnr": "6XJ4KP",
  "gdsPNR": null,
  "bookingToken": null,
  "surName": "Al Sayed",
  "sendEmailNotification": true,
  "bspCommissionType": null,
  "passengersSelectedAncillaries": {
    "PAX1": {
      "setSegmentsAncillaries": {
        "SEG1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      },
      "setJourneysAncillaries": {
        "J1": [
          { "additionalServiceCategory": "Baggage", "ancillaryCode": "20KG", "quantity": "1" }
        ]
      }
    },
    "PAX2": {
      "setSegmentsAncillaries": {},
      "setJourneysAncillaries": {}
    }
  }
}
```

### 7.4 Sample — BookAfterHold response

Full response in BookAfterHold-Response-Sample.json — identical seat/price data to Hold and Book for the same PNR, now with tickets issued.

## 8. RetrieveBooking

Looks up an existing PNR at any later point in its lifecycle.

### 8.1 Request — RetrieveBookingRequestDto (OrderBaseRequestDto)

No fields beyond the shared base in §5.1 — pnr (or gdsPNR/bookingToken) plus surName is enough to look a booking up.

### 8.2 Response — RetrieveBookingResponseDto

| Field | Type | Description / purpose |
|---|---|---|
| `supplier / voidCapability / tickets / creationDate / ticketingDate` | — | Same meaning as Book (§6.2). |
| `fareType / bookedCredentialsSelector / holdExpirationDate` | *enum? / enum? / DateTimeOffset?* | Null once ticketed; holdExpirationDate is only meaningful if the PNR is still on hold. |
| `passengersSelectedAncillaries` | *Dictionary<paxId, PassengerSelectedAncillary>* | Confirmed ancillary selections. |
| `passengersBookedSeats` | *Dictionary<paxId, Dictionary<segmentId, BookedSeatDetails>>* | Confirmed seat assignment — same shape throughout the lifecycle. |
| `bookingStatus` | *enum BookingStatus* | Confirmed / Cancelled / etc. |
| `pcc` | *string* | Pseudo city code, for GDS suppliers. |

| Field | Type | Description / purpose |
|---|---|---|
| `selectedOffer` | *RetrieveBookingOrderType* | Same passengerFareBreakdown shape as every other OrderType. |

### 8.3 Sample — RetrieveBooking request

```
{
  "supplier": "AirArabia",
  "credentialsSelector": "Default",
  "pnr": "6XJ4KP",
  "gdsPNR": null,
  "bookingToken": null,
  "surName": "Al Sayed",
  "sendEmailNotification": null
}
```

### 8.4 Sample — RetrieveBooking response

Full response in RetrieveBooking-Response-Sample.json — the same seat/price data as every prior step, however long after booking this is called.

## 9. Supporting and shared types

### 9.1 Money

| Field | Type | Description / purpose |
|---|---|---|
| `amount` | *decimal* | Numeric price amount. |
| `currency` | *string* | ISO currency code. |

### 9.2 SeatPriceDetails

| Field | Type | Description / purpose |
|---|---|---|
| `totalPrice` | *Money* | Total price of the seat. |
| `basePrice` | *Money* | Base fare component. |
| `taxesPrice` | *Money* | Tax/fee component. |

### 9.3 BookedSeatDetails (response side)

| Field | Type | Description / purpose |
|---|---|---|
| `cellCode` | *string* | Seat code, e.g. "12A". |
| `name` | *string?* | Display name for the seat. |
| `description` | *string?* | Free-text description. |
| `price` | *Money* | Price of this seat, matching the shape of SelectedAncillaryDetails.Price. |
| `cellCharacteristics` | *List<CellCharacteristicType>* | Window/aisle/exit-row etc.; empty when sourced from FareCheck's ancillary channel, populated when sourced from a confirmed booking. |

## 9.4 SetAncillaryDetails / SetPassengerAncillary (request side)

| Field | Type | Description / purpose |
|---|---|---|
| `additionalServiceCategory` | *enum AdditionalServiceCategory* | None / Other / Baggage / Meal / Seat. |
| `ancillaryCode` | *string* | The ancillary being requested — for Seat-category entries, this is the seat code. |
| `quantity` | *string?* | How many of this ancillary, usually "1". |

SetPassengerAncillary wraps these per segment/journey:

| Field | Type | Description / purpose |
|---|---|---|
| `setSegmentsAncillaries` | *Dictionary<segmentId, List<SetAncillaryDetails>>?* | Ancillaries requested per flight segment. |
| `setJourneysAncillaries` | *Dictionary<journeyId, List<SetAncillaryDetails>>?* | Ancillaries requested per journey (can span multiple segments, e.g. baggage). |

## 9.5 SelectedAncillaryDetails (response side)

| Field | Type | Description / purpose |
|---|---|---|
| `additionalServiceCategory / ancillaryCode / quantity` | — | Echo of the request fields above. |
| `price` | *Money* | Resolved price for this ancillary. |
| `description` | *string* | Resolved description for this ancillary. |

## 9.6 SetSeatDetails / SetPassengerSeatSelection (request side)

The dedicated seat-selection channel on FareCheck, Hold and Book requests (§3.1/4.1/6.1).

| Field | Type | Description / purpose |
|---|---|---|
| `cellCode` | *string?* | Seat code being requested, e.g. "12A" — matches CellDetails.CellCode from the SeatMap response. |
| `cellId` | *string?* | Supplier-internal seat identifier — matches CellDetails.CellID from the SeatMap response. |

SetPassengerSeatSelection wraps this per segment:

| Field | Type | Description / purpose |
|---|---|---|
| `setSegmentsSeats` | *Dictionary<segmentId, SetSeatDetails>?* | One seat per segment per passenger — no journey-level variant, since a passenger occupies exactly one seat per physical flight segment. |

## 9.7 PadisCode — full IATA PADIS code table

Every code CellCharacteristicType.PadisCode can take, mapped to the description PadisSeatCharacteristicHelper produces for it. Numeric codes are valid PADIS values; the C# enum member for code "17" is named _17 since identifiers can't start with a digit — the value on the wire is always the plain code without the underscore.

| PadisCode | Description |
|---|---|
| Unknown | Unspecified seat characteristic |
| 1 | Restricted seat - General |
| 2 | Leg rest available |
| 3 | Individual video screen - Choice of movies |
| 4 | Not a window seat |
| 6 | Near galley seat |
| 7 | Near toilet seat |
| 8 | No seat at this location |
| 9 | Center seat (not window, not aisle) |
| 10 | Seat designated for RBD "A" |
| 11 | Seat designated for RBD "B" |
| 12 | Seat designated for RBD "C" |
| 13 | Seat designated for RBD "D" |
| 14 | Seat designated for RBD "F" |
| 15 | Seat designated for RBD "H" |
| 16 | Seat designated for RBD "J" |
| 17 | Seat designated for RBD "K" |
| 18 | Seat designated for RBD "L" |
| 19 | Seat designated for RBD "M" |
| 20 | Seat designated for RBD "P" |
| 21 | Seat designated for RBD "Q" |
| 22 | Seat designated for RBD "R" |
| 23 | Seat designated for RBD "S" |
| 24 | Seat designated for RBD "T" |
| 25 | Seat designated for RBD "V" |
| 26 | Seat designated for RBD "W" |
| 27 | Seat designated for RBD "Y" |
| 28 | Not fitted |
| A | Aisle seat |
| AA | All available aisle seats |
| AB | Seat adjacent to bar |
| AC | Seat adjacent to - closet |
| AG | Seat adjacent to galley |
| AJ | Adjacent aisle seats |
| AL | Seat adjacent to lavatory |

| PadisCode | Description |
|---|---|
| AM | Individual movie screen - No choice of movie selection |
| AR | No seat - airphone |
| AS | Individual airphone |
| AT | Seat adjacent to table |
| AU | Seat adjacent to stairs to upper deck |
| AV | Only available seats |
| AW | All available window seats |
| B | Seat with bassinet facility |
| BA | No seat - bar |
| BK | Blocked Seat for preferred passenger in adjacent seat |
| C | Crew seat |
| CC | Center section seat(s) |
| CH | Chargeable seat |
| CL | No seat - closet |
| CS | Conditional seat-contact airline |
| D | No seat - exit door |
| DE | Deportee |
| E | Exit and emergency exit |
| EA | Not on exit seat |
| EC | Electronic connection for lap top or FAX machine |
| EK | Economy comfort seat |
| EX | No seat - emergency Exit |
| F | Added seat |
| FC | Front of cabin class/compartment |
| G | Seat at forward end of cabin |
| GF | General facility |
| GN | No seat - galley |
| GR | Group seat - offered to travellers belonging to a group |
| H | Seat with facilities for handicapped/incapacitated passenger |
| I | Seat suitable for adult with an infant |
| IA | Inside aisle seats |
| IE | Seat not suitable for child |
| J | Rear facing seat |
| K | Bulkhead seat |
| KA | Bulkhead seat with movie screen |

| PadisCode | Description |
|---|---|
| KN | Bulkhead, no seat |
| L | Leg space seat |
| LA | No seat – lavatory |
| LB | Rear facing lie flat seat |
| LE | Left facing lie flat seat more left angle |
| LF | Lie flat seat |
| LG | No seat - luggage storage |
| LH | Restricted seat - offered on long-haul segments |
| LL | Left facing lie flat seat |
| LR | Right facing lie flat seat |
| LS | Left side of aircraft |
| LT | Right facing lie flat seat more right angle |
| M | Seat without a movie view |
| MA | Medically OK to travel |
| MS | Middle seat |
| N | No smoking seat |
| O | Preferential seat |
| OW | Overwing seat(s) |
| P | Extra seat for comfort - arm rest can be raised |
| PC | Pet cabin |
| Q | Seat in a quiet zone |
| RS | Right side of aircraft |
| S | Smoking seat |
| SC | Skycouch |
| SO | No seat - storage space |
| ST | No seat - stairs to upper deck |
| T | Rear/Tail section of aircraft |
| TA | No seat - table |
| U | Seat suitable for unaccompanied minors |
| UP | Upper deck |
| V | Seat to be left vacant or offered last |
| W | Window seat |
| WA | Window and aisle together |
| X | No facility seat (indifferent seat) |
| Z | Buffer zone seat |

| PadisCode | Description |
|---|---|
| 1A | Seat not allowed for infant |
| 1B | Seat not allowed for medical |
| 1C | Seat not allowed for unaccompanied minor |
| 1D | Restricted recline seat |
| 1M | Seat with movie view |
| 1W | Window seat without window |
| 3A | Individual video screen - No choice of movie |
| 3B | Individual video screen-Choice of movies, games, information, etc |
| 6A | In front of galley seat |
| 6B | Behind galley seat |
| 7A | In front of toilet seat |
| 7B | Behind toilet seat |
| 70 | Individual video screen - services unspecified |
| 71 | No seat --access to handicapped lavatory |
| 72 | Undesirable seat |

### 9.8 Other enums

| Enum | Values |
|---|---|
| DeckType | Unknown / Main / Upper / Lower |
| CellCategory | Seat / Aisle / Lavatory / Other |
| CellAvailability | Blocked / Occupied / AvailableFree / AvailablePaid / AvailableAferReservation |
| AdditionalServiceCategory | None / Other / Baggage / Meal / Seat |

## 10. End-to-end flow

Every sample in this document uses the same PNR, passengers and seats, so the seat codes and prices trace straight through from shopping to retrieval:

| Field | Type | Description / purpose |
|---|---|---|
| 1. GET SeatMap | *§2* | Client sees seat 12A (25 AED) and 12B (15 AED) available in the exit row. |
| 2. FareCheck | *§3* | Client re-prices with those two seats selected; paxTotalSeatAmount reflects 25 / 15 AED. No PNR yet. |
| 3. Hold | *§4* | PNR 6XJ4KP created; passengersBookedSeats and the fare breakdown now read from the confirmed hold. |
| 4. FareConfirmAfterHold | *§5* | Re-confirms the same PNR's price — identical seat data, copied from RetrieveBooking. |

| Field | Type | Description / purpose |
|---|---|---|
| `5. Book or BookAfterHold` | *§6 / §7* | PNR 6XJ4KP is ticketed — seat assignment unchanged, tickets added. |
| `6. RetrieveBooking` | *§8* | Looking up 6XJ4KP later returns the exact same seat data as every prior step. |